© 2024 IEEE. Personal use of this material is permitted. Permission from IEEE must be obtained for all other uses, in any current or future media, including reprinting/ republishing this material for advertising or promotional purposes, creating new collective works, for resale or redistribution to servers or lists, or reuse of any copyrighted component of this work in other works. Accepted manuscript of: Y. Lu, S. Yao, Y. Wang, H. Wang, and G. Chen, "A Geometric Approach for the Bifurcation Modeling and Analysis of Plane-Symmetric Bricard Mechanisms," 2024 6th International Conference on Reconfigurable Mechanisms and Robots (ReMAR), Chicago, IL, USA, 2024, pp. 180–185. DOI: 10.1109/ReMAR61031.2024.10617685

A Geometric Approach for the Bifurcation Modeling and Analysis of Plane-symmetric Bricard Mechanisms

Yuliang Lu[1], Siyue Yao[1], Yanjun Wang[2], Hao Wang[2], and Genliang Chen[3]

***Abstract*— As a classical type of overconstrained linkages, the spatial 6R single-loop Bricard mechanisms have long attracted researchers' attention. In our previous work, a comprehensive geometric approach is established for the kinematics analysis of general plane-symmetric Bricard mechanisms, with an intuitive interpretation of kinematic properties. In this paper, this novel approach is further extended for the bifurcation analysis of the plane-symmetric Bricard mechanisms. Owing to the geometric representation, the condition of bifurcations can be interpreted as a continuous intersection of two coplanar ellipses, including their degeneration. All possible scenarios are analyzed in detail, based on which the corresponding kinematic conditions can be specified to distinguish Bricard mechanisms with bifurcations. A comprehensive categorization is provided to classify the candidates into different categories according to various conditions. Case studies are made to enumerate all categories of kinematic bifurcations, in an intuitive way, with the geometric illustration of mobility in these special configurations.**

I. INTRODUCTION

The Bricard mechanisms, as one of the major categories of single-loop overconstrained linkages, was first proposed and classified by Bricard [1] in the 1890s. Owing to various symmetric properties, these 6R close-loop mechanisms still possess mobility while disobeying the Grübler-Kutzbach formula. Among them, plane-symmetric ones have been extensively studied for their potential applications in different fields, such as deployable structures [2], underwater graspers [3], and compliant mechanisms [4].

Many researchers have designed diverse plane-symmetric Bricard mechanisms with special kinematic properties. Some of them [5]–[8] put forward various plane-symmetric Bricard mechanisms capable of deploying from a compact bundle to a planar configuration, including the triangular, rectangular, trapezoidal and hexagonal cases. Qi et al. [9] developed the retractable Bricard linkage into a class of large deployable mechanisms through ingenious assembly. Zhang et al. [10] used the fourfold-symmetric linkages as a gripper and analyzed the motion properties of the structure. Wang et al. [11] presented a class of Bricard mechanisms with perpendicular joint axes. Dai et al. [12] proposed a plane-symmetric Bricard linkage possessing line-symmetric bifurcations. As well, some researchers have designed complex linkages with plane-symmetric Bricard bifurcations. For instance, Wang et al. [13] and Tang et al. [14] designed different eight-bar linkages with reconfiguration to plane-symmetric Bricard.

Notwithstanding the multifarious mechanisms with satisfying properties, the plane-symmetric Bricard mechanism stands a chance of complicated bifurcations on account of the symmetry property. Such features should be avoided when using the Bricard linkages as a one-DOF mechanism, such as deployable structures, but it may be valuable for designing a reconfigurable mechanism. Chen et al. [15] studied the kinematics and bifurcation of a kind of line and plane symmetric Bricard mechanism. Zhang et al. [16] presented a double-spherical 6R linkage and investigated its bifurcation and trifurcation based on screw theory. A type of two-fold symmetrical 6R foldable frame is discovered by Chen and You [17], showing bifurcations during deployment. Husty et al. [18] solved the Bennett linkage using kinematic mapping. Except for these specific cases, researchers have made efforts on the explanations of reconfigurations. Dai et al. [19] constructed 3D surfaces to represent motion spaces of adjacent joints and the intersection of the toroids exhibited the motion features and singularities. Feng et al. [20] made a thorough bifurcation study of general plane-symmetric Bricard linkage through solving DH matrix equations. Both approaches demand rather complicated calculations while some parameter requirements are not clear enough.

To this end, this paper aims to search for plane-symmetric Bricard mechanisms with bifurcations rapidly and give a visualized explanation of bifurcation and corresponding detailed constraints of the geometry parameters. In our prior work [21], a geometric method analyzing the motion properties of general plane-symmetric Bricard mechanisms has been proposed which equalized the solution of the configuration to the intersection of two ellipses. This paper indicates that the approach is capable of guiding the design of plane-symmetric Bricard mechanisms with specific properties.

This paper is organized as follows: In Sec. II, the procedures and theories of the proposed method are simply introduced. The degeneration conditions of the ellipses and the classification of bifurcations are exhibited. Section III gives detailed constraints and lists examples for each type of reconfiguration. In Sec. IV, some missing bifurcations which may cause the mechanism asymmetrical motion are supplemented. Finally, conclusions are drawn in Sec. V.

* This work is supported in partial by the National Key R&D Program of China under Grant 2022YFB4701204, the Natural Science Foundation of China under Grant 52022056 and the Science and Technology Commission of Shanghai Municipality under Grant 21NL2600200. Corresponding author: Genliang Chen.

[1] Y. Lu and S. Yao are with the Shanghai Key Laboratory of Digital Manufacturing for Thin-Walled Structures, and the Meta Robotics Institute, Shanghai Jiao Tong University (SJTU), Shanghai, 200240, P.R. China. yulianglu@sjtu.edu.cn and sy.yao@sjtu.edu.cn

[2] Y. Wang and H. Wang are with the Institute of Marine Equipment, SJTU, Shanghai, 200240, P.R. China. yanjunwang@sjtu.edu.cn and wanghao@sjtu.edu.cn

[3] G. Chen is with the State Key Laboratory of Mechanical Systems and Vibration, and the Meta Robotics Institute, SJTU, Shanghai, 200240, P.R. China. leungchan@sjtu.edu.cn

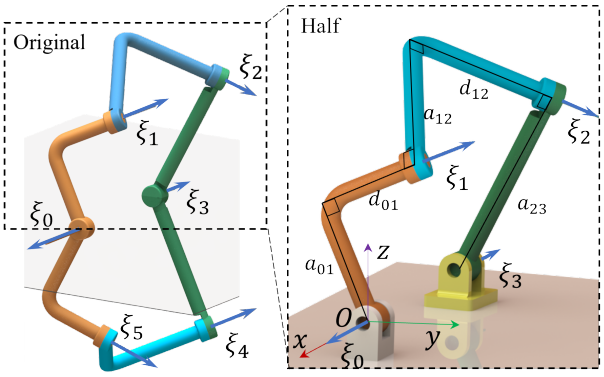


Fig. 1. Construction of the half Bricard mechanism

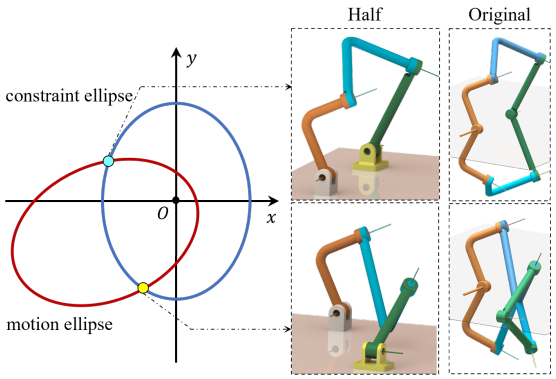


Fig. 2. Identification of the configuration

II. Bifurcation Modeling of Plane-Symmetric Bricard Mechanisms

This section presents a brief introduction to the mentioned method equalizing the configuration solution to the intersection of two ellipses. Based on the degeneration and intersection condition of the two ellipses, classification of probable bifurcation of the Bricard mechanism is conducted.

A. An Introduction to the Geometric Approach

In our previous work [21], a novel approach to the kinematics of general plane-symmetric Bricard mechanisms was proposed. The detailed procedure is as follows:

1) Construction of the half Bricard linkage:

As shown in Fig. 1, the original complete plane-symmetric Bricard linkage is split about its symmetry plane to construct the corresponding half Bricard linkage as a spatial 'RRRRE' linkage with one DOF. Some of the geometric parameters are shown in the figure, all presented in DH parameters. a refers to the link length and d refers to the link offset. α and θ which refer to link twists and joint angles are not marked in the figure.

2) Construction and intersection of the constraint and motion ellipses:

Based on the axial pose of the intermediate joints of the half Bricard linkage, we turned the solution of configuration into the intersection of two coplanar ellipses, which is a completely solved problem. The ellipses can be obtained according to the geometric parameters of the linkage and the given input of the active joint ξ_0. In detail, the two ellipses, named constraint ellipse and motion ellipse, are constructed from the following equations regardless of degeneration.

Constraint ellipse (con. ellip. as abbreviation):

$$\frac{x^2}{a_{23}^2}+\frac{y^2}{\sin^2\alpha_{23}}=1 \tag{1}$$

Motion ellipse (mot. ellip. as abbreviation):

$$\frac{(x-d_{12}y-h_1)^2}{(a_{12}\cos\delta)^2}+\frac{(y-\cos\alpha_{12}\sin\delta)^2}{(\sin\alpha_{12}\cos\delta)^2}=1 \tag{2}$$

in which $h_1=-\sin\theta_0 a_{01}+\cos\theta_0 d_{01}\sin\alpha_{01}$ and $\sin\delta=\cos\theta_0\sin\alpha_{01}$ are known parameters with given input. All the parameters including x and y have actual geometric meanings which have been discussed in our previous work.

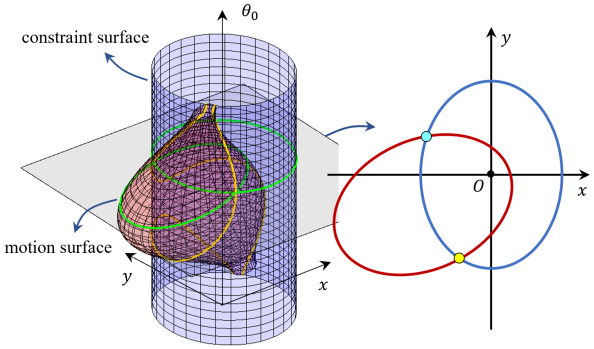


Fig. 3. Construction of constraint and motion surfaces

3) Identification of the configuration:

Shown in Fig. 2, as the two ellipses represent the rotation of joints in the half Bricard mechanism, every intersection point stands for a distinct set of joint angles under the given input, leading to a distinct configuration. Through mirroring the half Bricard mechanism, the corresponding configuration of the original mechanism can be identified.

4) Intuitive exhibition of kinematic characteristics:

A pencil of ellipses will be generated along with different inputs of the active joint. With continuous input, the ellipses change accordingly. As shown in Fig. 3, the ellipses are expanded into two surfaces by extending the $\{x,y\}$ plane with the active input θ_0. Each layer of the 3D graphic represents the configuration situation under a given input. Furthermore, the intersection line of two surfaces manifests the kinematic properties of the mechanism, such as the dead center configurations and the branches of continuous motion, in an intuitive way.

The geometric method obtains configurations of general plane-symmetric Bricard mechanisms in a straightforward procedure. Compared to solving equations based on DH matrices or screw theory, this approach shows an advantage in simplicity and intuition of kinematic properties. Meanwhile, through designing geometric characteristics of the intersection points, such as tangency and continuity, Bricard mechanisms with special motion properties can be constructed.

B. Geometric Interpretation and Categorization of Bifurcation

As discussed above, every intersection point of the two ellipses represents a distinct configuration under the given input. That is to say, if a plane-symmetric Bricard mechanism is still deformable with a fixed input, which means the mechanism has bifurcations, the ellipses under this actuation angle should have continuous intersection points.

However, the denominators in Eq. (1) and Eq. (2) may turn to zero in special cases where some adjacent joint axes of the half Bricard mechanism are parallel or intersecting and the corresponding ellipses degenerate into segments or even points. Continuous intersection of the degenerated shapes should also be considered. Table I lists all the possible degeneration conditions.

Nevertheless, the first two cases of point degeneration with collinear axes caused one of the related axes redundant, then the corresponding complete Bricard mechanism should not be regarded as a 6R mechanism. Thus, these two degeneration cases are not taken into account in the following parts. Besides, the case $\cos\delta = 0$ means that the joint axis ξ_1 is perpendicular to the symmetry plane, it only occurs in a particular input during the activation.

Table II presents all the possibilities that constraint and motion ellipses continuously intersect. The only way for two ellipses to share infinite intersection points is that they coincide, while there is no chance that an ellipse and a segment may intersect continuously. Point-on-ellipse and point-on-segment cases are listed in the table though they seemingly exhibit only one intersection point. This is because the point is regarded as a whole ellipse with major and minor axes equal to zero. In terms of geometric representation, the point contains the full turn of ξ_1. Detailed information of each case is discussed in the following section.

III. Case Study of Plane-Symmetric Bricard Mechanisms with Bifurcations

From the perspective of elliptical intersection and degeneration, the bifurcations of plane-symmetric Bricard mechanisms can be divided into four main categories. Some of them will be further subdivided on the basis of geometric constraints and examples are given for every category.

TABLE I

Degeneration of Constraint and Motion Ellipses

Shape	Parameter	Geometry	Con. Ellip.	Mot. Ellip.
segment	$a_{23}=0$	ξ_2,ξ_3intersect	$x=0$	/
	$\sin\alpha_{23}=0$	ξ_2,ξ_3parallel	$y=0$	/
	$a_{12}=0$	ξ_1,ξ_2intersect	/	$x-d_{12}y-h_1=0$
	$\sin\alpha_{12}=0$	ξ_1,ξ_2parallel	/	$y=\cos\alpha_{12}\sin\delta$
point	$a_{23}=\sin\alpha_{23}=0$	ξ_2,ξ_3collinear	$(0;0)$	/
	$a_{12}=\sin\alpha_{12}=0$	ξ_1,ξ_2collinear	/	$\begin{pmatrix} d_{12}\cos\alpha_{12}\sin\delta+h_1 \\ \cos\alpha_{12}\sin\delta \end{pmatrix}$
	$\cos\delta=0$	$\xi_1$vertical		

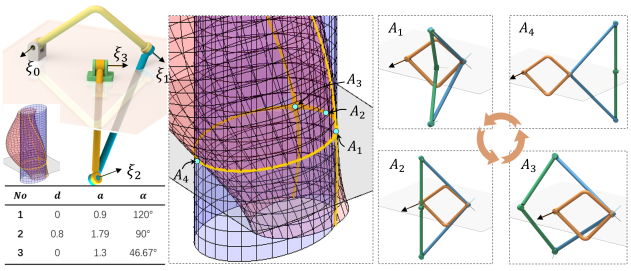


Fig. 4. Bifurcation from Bricard to Myard under coincident case.

A. Coincidence of Motion and Constraint Ellipses

It is clear that the coincidence of two ellipses represents the selfsame equation parameters of Eq. (1) and (2), as shown in Eq. (3).

$$\begin{cases} d_{12}=0 \\ h_1=0 \\ \cos\alpha_{12}\sin\delta=0 \\ a_{12}\cos\delta=\pm a_{23} \\ \sin\alpha_{12}\cos\delta=\pm\sin\alpha_{23} \end{cases} \tag{3}$$

A classification discussion is conducted based on the third constraint $\cos\alpha_{12}\sin\delta=0$.

1) $\alpha_{12}=\frac{\pi}{2},\frac{3\pi}{2}$*:* After eliminating the input angle from the original equation, the constraints the DH parameters should satisfy to achieve such bifurcation are shown in Eq. (4) and (5).

$$\begin{cases} d_{12}=0 \\ \alpha_{12}=\frac{\pi}{2},\frac{3\pi}{2} \\ a_{12}\sin\alpha_{23}=\pm a_{23} \end{cases} \tag{4}$$

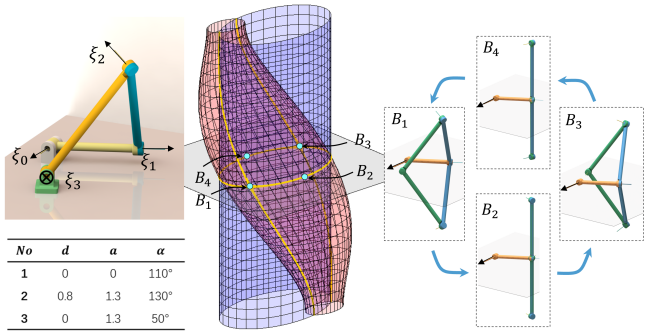
$$\cos^2\alpha_{23}(a_{01}^2+d_{01}^2\sin^2\alpha_{01})=\sin^2\alpha_{01}a_{01}^2 \tag{5}$$

Geometrically speaking, Eq. (4) points out the relationship of the rest movable parts except the fixed actuation joint ξ_0, which indicates a bifurcation from plane-symmetric Bricard mechanism to Myard 5R mechanism [22], a symmetric single-loop 5R linkage. In the meantime, Eq. (5) guarantees assemblability of the whole mechanism. Fig. 4 gives an example as a direct exhibition. The corresponding kinematic parameters in DH convention are provided, according to which the surfaces of constraint and motion ellipses can

TABLE II

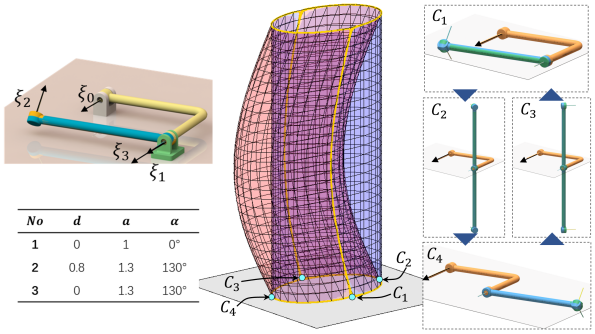
Classification of Continuous Intersection

		Mot. Ellip.		
	Shape	ellipse	segment	point
Con. Ellip.	ellipse	coincide	/	on
	segment	/	collinear	on



No	d	a	α
1	0	0	110°
2	0.8	1.3	130°
3	0	1.3	50°

Fig. 5. Bifurcation from Bricard to Bennett under coincident case



No	d	a	α
1	0	1	0°
2	0.8	1.3	130°
3	0	1.3	130°

Fig. 6. Bifurcation from Bricard to revolute joint under coincident case

be obtained directly. For the surfaces may be illegible, both the complete surfaces and coincident parts are exhibited. As shown in Fig. 4, under most circumstances, there are only two intersection points on a layer, meaning two different configurations under the same input, but when it comes to $\theta_0 = 37.59°$, the two ellipses completely overlap. For instance, a configuration of the half Bricard mechanism on the coincidence layer is shown in Fig. 4, and four different configurations of the whole mechanism are given to demonstrate motion of the local Myard mechanism. Besides, it is notable that on the coincidence layer, only under A_1 and A_3 configurations can the active joint ξ_0 move since other points lack connected intersection points on the upper or lower layers.

2) $\delta = 0, \pi$: Under this circumstance, a straightforward conclusion can be given that when the two ellipses coincide, joint axis ξ_1 is right on the symmetry plane, because h_1 and $\sin\delta$ are all zero. Further, based on the symmetry of the whole mechanism, ξ_1 and ξ_5 also coincide. Similar to the former case, Eq. (6) gives the geometric constraints of the movable part and either of the constraints in Eq. (7) assures the possibility for ξ_1 on the plane.

$$\begin{cases} a_{12} = a_{23} \\ \alpha_{12} = \pm\alpha_{23} + k\pi, k \in \mathbb{Z} \end{cases} \tag{6}$$

$$a_{01} = 0 \quad or \quad \alpha_{01} = 0, \pi \tag{7}$$

An example is shown in Fig. 5 when $\alpha_{12} = -\alpha_{23} + k\pi$, in which case the last four poles construct a symmetric Bennett linkage [23]. During the traversal of points on the coinciding ellipses, the local Bennett mechanism has a special configuration as a straight line at the points B_2 and B_4. Also, the actuation joint is movable only when the mechanism is on the status of B_1 and B_3.

Another example shown in Fig. 6 sheds light on the constraint $\alpha_{12} = \alpha_{23} + k\pi$, causing the overlapping of the last two poles of the half Bricard mechanism. When the constraint and motion ellipse coincide, ξ_1 and ξ_3 are collinear, leading to the mechanism's degeneration to a single revolute joint.

B. Collinearity of Line Segments

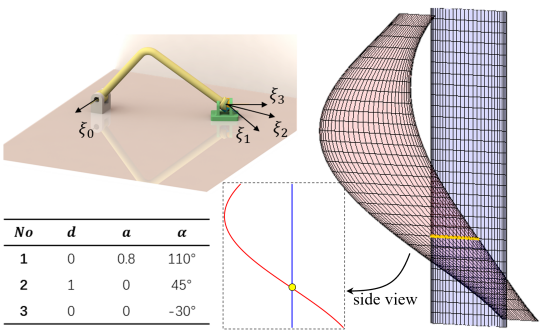
When both ellipses are degenerated into line segments, they intersect continuously only if they are collinear. Unlike ellipses which have to coincide, segments can partially overlap to reach a bifurcation, which represents a partial rotation of ξ_1. As the constraint ellipse can only degenerate to segments on the coordinate axes, the conditions that two segments overlap can generally divided into two types.

1) segments on $x = 0$: For the two ellipses degenerated into segments on $x = 0$, basic constraints are shown in Eq. (8). The constraint $h_1 = 0$ actually infers that in most cases with general DH parameters, there is only one layer of the constraint and motion surfaces on which segments are collinear. Further, though the segments are collinear, to make sure that they overlap, the Inequation (9) should be satisfied for the mechanism to successfully assemble.

$$\begin{cases} d_{12} = 0 \\ a_{12} = 0 \\ a_{23} = 0 \\ h_1 = 0 \end{cases} \tag{8}$$

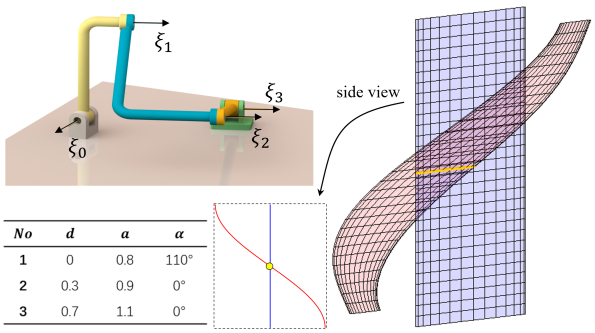
$$|\sin\alpha_{23}| > |\sin\delta\cos\alpha_{12}| - |\cos\delta\sin\alpha_{12}| \tag{9}$$

Geometrically speaking, as an example shown in Fig. 7, when the segments are collinear, the three joint axes ξ_1, ξ_2 and ξ_3 intersect at the same point, and construct a 4R spherical linkage together with the revolute axis in the planar pair, leading to the mobility. A side view of the constraint and motion surfaces is given in case the spatial relationship between the two surfaces is illegible under degeneration cases. After mirroring the half mechanism comes a 5R



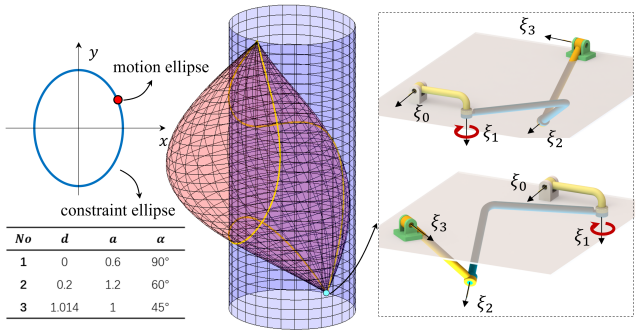
No	d	a	α
1	0	0.8	110°
2	1	0	45°
3	0	0	-30°

Fig. 7. Degeneration from Bricard to spherical 5R under collinear case



No	d	a	α
1	0	0.8	110°
2	0.3	0.9	0°
3	0.7	1.1	0°

Fig. 8. Degeneration from Bricard to planar 5R under collinear case



No	d	a	α
1	0	0.6	90°
2	0.2	1.2	60°
3	1.014	1	45°

Fig. 9. Bifurcation from Bricard to revolute joint with point degeneration

spherical linkage consisting of $\xi_1 \sim \xi_5$. However, it is worth mentioning that under such circumstance, the mobility of the mechanism can hardly be called a bifurcation as the linkage can only be assembled under the fixed input. The only way for ξ_0 to be movable is that the axis also passes through the copoint, leading to a 5R spherical linkage in the half Bricard mechanism, and 6R spherical as a whole.

2) segments on $y = 0$:

For segments on $y = 0$, constraints of parameters are presented in Eq. (10), where $\sin\delta = 0$ represents that in most cases, only when $\theta_0 = \frac{\pi}{2}$ can the segments be collinear. In order to make the segments overlap, Inequation (11) should be taken into consideration.

$$\begin{cases} \alpha_{12} = 0, \pi \\ \alpha_{23} = 0, \pi \\ \sin\delta = 0 \end{cases} \tag{10}$$

$$a_{01} < a_{12} + a_{23} \tag{11}$$

Fig. 8 shows that the joint axes ξ_1, ξ_2 and ξ_3 are parallel and form an 'RRRP' one-DOF planar linkage with the planar pair when two segments overlap. For the whole original Bricard mechanism, it degenerates into a 5R planar linkage with $\xi_1 \sim \xi_5$ all parallel. Similarly, ξ_0 is also forcibly fixed in most cases, and it is movable only when ξ_0 is also parallel to the other three axes, constructing an 'RRRRP' planar linkage as a half mechanism and a 6R planar linkage after mirroring.

C. Intersection of Point and Ellipse/Line Segment

As aforementioned, the only point degeneration case worthy of analyzing is $\cos\delta = 0$. The constraints of geometric parameters leading to point-and-ellipse intersection are listed as Eq. (12) and (13). Eq. (12) is a must for the degeneration from an ellipse to a point, while Eq. (13) in elliptic equation form ensures the intersection of the degenerated point and the ellipse.

$$\alpha_{01} = \frac{\pi}{2}, \frac{3\pi}{2} \tag{12}$$

$$\frac{(d_{12}\cos\alpha_{12} + d_{01})^2}{a_{23}^2} + \frac{\cos^2\alpha_{12}}{\sin^2\alpha_{23}} = 1 \tag{13}$$

As an example shown in Fig. 9, when the motion ellipse degenerates into a point, joint axis ξ_1 is perpendicular to the symmetry plane, leading to a full turn around the axis with unchanged spatial relationship between ξ_2 and ξ_3. Moreover, similar to the cases of ellipse coincidence, only at the two configurations shown in the figure can the mechanism move along the two intersecting curves of constraint and motion surfaces. In the perspective of the complete plane-symmetric Bricard mechanism, ξ_1 and ξ_5 are collinear, so the distal parts of the mechanism can rotate around ξ_1, degenerating into a revolute joint.

This section also covers the case where the degenerated point intersects with a segment because it is just a degeneration of the mentioned Eq. (13) in elliptic equation form. The constraints are given in Eq. (14). The kinematic properties on the continuous intersection layer are almost the same with a full turn of ξ_1. The only difference lies in the chance that there may exist more than two configurations at which the active joint can move.

$$\begin{cases} \alpha_{01} = \frac{\pi}{2}, \frac{3\pi}{2} \\ a_{23} = 0 \\ d_{12}\cos\alpha_{12} + d_{01} = 0 \\ |\cos\alpha_{12}| \le |\sin\alpha_{23}| \end{cases} \quad or \quad \begin{cases} \alpha_{01} = \frac{\pi}{2}, \frac{3\pi}{2} \\ \alpha_{23} = 0, \pi \\ \alpha_{12} = \frac{\pi}{2}, \frac{3\pi}{2} \\ |d_{01}| \le a_{23} \end{cases} \tag{14}$$

As all the bifurcation categories mentioned have been discussed detailedly, a comprehensive overlook on the types of continuous intersection and corresponding bifurcated mechanisms can be concluded, as shown in Table III.

TABLE III
CONTINUOUS INTERSECTION AND CORRESPONDING MECHANISM

Continuous Intersection	Axis Feature	Mechanism Type
ellipse coincident	$\xi_1 \perp \xi_2$	Myard 5R
	$\xi_1 \in$sym. plane	Bennett 4R
		R
segments collinear	$\xi_1 \sim \xi_5$copoint	spherical 5R
	$\xi_1 \sim \xi_5$parallel	planar 5R
point on ellipse	$\xi_1 \perp$sym. plane	R
point on segment		

TABLE IV

BIFURCATIONS FROM SYMMETRICAL TO ASYMMETRICAL MOTIONS

Parameter Constraints	Axis Feature	Mechanism Type
$a_{01}=0 \quad or \quad \alpha_{01}=0,\pi$	$\xi_1,\xi_5 \in$ sym. plane	R
$a_{12}=0$	ξ_1,ξ_2,ξ_4,ξ_5copoint	spherical 4R
$\alpha_{12}=0,\pi$	ξ_1,ξ_2,ξ_4,ξ_5parallel	planar 4R

IV. DISCUSSION

In this section, some bifurcations causing asymmetrical motion of the mechanism are discussed with simplified constraints on the geometric parameters.

The bifurcations analysis in plane-symmetric Bricard mechanisms in this paper is based on the theory utilizing the half Bricard mechanism. Thus, the bifurcation cases found actually belong to the half mechanism, forcing the mirrored complete mechanism to move symmetrically, though the original mechanism may have the ability to move both asymmetrically and symmetrically. In other words, the constraints given above may be suitable for a plane-symmetric bifurcation, but too strict for a general bifurcation. To be specific, the cases under ellipse coincidence situation, in which ξ_1 is right on the symmetry plane, show the capability of the distal parts of the complete mechanism to rotate around the coincident ξ_1 and ξ_5, constructing an asymmetric configuration with fixed relative pose between ξ_2 and ξ_3. In addition, clearly both the spherical 5R and planar 5R linkages have two DOFs, leading to not only symmetrical configurations.

Table IV gives the simplified constraints for the Bricard mechanism to bifurcate from symmetric configurations to asymmetric ones. The required parameter constraints are all part of the corresponding bifurcation cases mentioned above, and so are the features of axes. In these cases, at the bifurcation configuration, the original mechanism can move with fixed input, but in an asymmetric way.

V. CONCLUSIONS

In this paper, a geometric approach is employed to analyze the bifurcations of plane-symmetric Bricard mechanisms. Based on line symmetry, the method turns the solving of configurations into the intersection of two coplanar ellipses constructed from the geometric parameters of the mechanism. In the opposite direction, Bricard mechanisms with special kinematic properties can be designed through specific arrangement of the geometric relationship of the ellipses. With the idea of continuous intersection, the bifurcation cases are listed and further divided into six corresponding bifurcated mechanism types. Examples are intuitively provided for every category. As supplementary, some asymmetric bifurcations are provided by simplifying the geometric constraints of the former cases. This paper verifies the proposed method to be an effective way to construct plane-symmetric Bricard mechanisms with specific characteristics.

REFERENCES

[1] R. Bricard, "Mémoire sur la théorie de l'octaèdre articulé," *Journal de Mathématiques pures et appliquées*, vol. 3, pp. 113–148, 1897.

[2] X. Qi, H. Huang, B. Li, and Z. Deng, "A large ring deployable mechanism for space satellite antenna," *Aerospace Science & Technology*, vol. 58, pp. 498–510, 2016.

[3] E. Zhi, B. Phillips, K. Becker, G. Whittredge, and R. Wood, "Rotary-actuated folding polyhedrons for midwater investigation of delicate marine organisms," *Science Robotics*, vol. 3, no. 20, p. eaat5276, 2018.

[4] G. Chen, Y. Chai, S. Tang, Y. Chen, L. Kong, and H. Wang, "A study on the flexible Bricard mechanisms with plane-symmetric structure," *Mechanism and Machine Theory*, vol. 188, p. 105404, 2023.

[5] Y. Chen, Z. You, and T. Tarnai, "Threefold-symmetric Bricard linkages for deployable structures," *International journal of solids and structures*, vol. 42, no. 8, pp. 2287–2301, 2005.

[6] A. D. Viquerat, T. Hutt, and S. D. Guest, "A plane symmetric 6R foldable ring," *Mechanism and Machine Theory*, vol. 63, pp. 73–88, 2013.

[7] S. Lyu, P. Yao, H. Xiao, W. Zhang, and X. Ding, "Approximating cylinders with bundle-folding plane-symmetric Bricard linkages," *International Journal of Mechanical Sciences*, vol. 221, p. 107231, 2022.

[8] F. Yang, Z. You, and Y. Chen, "Foldable hexagonal structures based on the threefold-symmetric Bricard linkage," *Journal of Mechanisms and Robotics*, vol. 12, no. 1, p. 011012, 10 2019.

[9] X. Qi, H. Huang, Z. Miao, B. Li, and Z. Deng, "Design and mobility analysis of large deployable mechanisms based on plane-symmetric Bricard linkage," *Journal of Mechanical Design*, vol. 139, no. 2, p. 022302, 11 2016.

[10] H. Zhang, X. Zhang, B. Zhu, R. Wang, and Q. Chen, "Multistability analysis for the compliant four-fold Bricard loops," in *2018 International Conference on Manipulation, Automation and Robotics at Small Scales (MARSS)*, 2018, pp. 1–6.

[11] J. Wang and X. Kong, "A novel method for constructing multimode deployable polyhedron mechanisms using symmetric spatial compositional units," *Journal of Mechanisms and Robotics*, vol. 11, no. 2, 2019.

[12] K. Zhang and J. S. Dai, "Geometric constraints and motion branch variations for reconfiguration of single-loop linkages with mobility one," *Mechanism and Machine Theory*, vol. 106, pp. 16–29, 2016.

[13] R. Wang, Y. Song, and J. S. Dai, "Reconfigurability of the origami-inspired integrated 8R kinematotropic metamorphic mechanism and its evolved 6R and 4R mechanisms," *Mechanism and Machine Theory*, vol. 161, p. 104245, 2021.

[14] Z. Tang and J. S. Dai, "Bifurcated configurations and their variations of an 8-bar linkage derived from an 8-kaleidocycle," *Mechanism and Machine Theory*, vol. 121, pp. 745–754, 2018.

[15] Y. Chen and W. H. Chai, "Bifurcation of a special line and plane symmetric Bricard linkage," *Mechanism and Machine Theory*, vol. 46, no. 4, pp. 515–533, 2011.

[16] K. Zhang and J. S. Dai, "Reconfiguration of the plane-symmetric double-spherical 6R linkage with bifurcation and trifurcation," *Proceedings of the Institution of Mechanical Engineers, Part C: Journal of Mechanical Engineering Science*, vol. 230, no. 3, pp. 473–482, 2016.

[17] Y. Chen and Z. You, "Two-fold symmetrical 6R foldable frame and its bifurcations," *International Journal of Solids and Structures*, vol. 46, no. 25, pp. 4504–4514, 2009.

[18] K. Brunnthaler, H.-P. Schröcker, and M. Husty, "A new method for the synthesis of Bennett mechanisms," *Proceedings of CK2005, Cassino*, 2005.

[19] P. C. López-Custodio, J. S. Dai, and J. M. Rico, "Branch reconfiguration of Bricard linkages based on toroids intersections: plane-symmetric case," *Journal of Mechanisms and Robotics*, vol. 10, no. 3, 03 2018.

[20] H. Feng, Y. Chen, J. Dai, and G. Gogu, "Kinematic study of the general plane-symmetric Bricard linkage and its bifurcation variations," *Mechanism and Machine Theory*, vol. 116, pp. 89–104, 2017.

[21] S. Yao, Y. Lu, X. Zhu, T. Yan, Y. Yang, H. Wang, and G. Chen, "On the kinematics of general plane-symmetric Bricard mechanisms," *Mechanism and Machine Theory*, vol. 190, p. 105433, 2023.

[22] F. Myard, "Contribution à la géométrie des systèmes articulés," *Bulletin de la Société Mathématique de France*, vol. 59, pp. 183–210, 1931.

[23] G. T. Bennett, "A new mechanism," *Engineering*, vol. 76, p. 777, 1903.